



OARS AND OARSMEN



◀ PILE OF OARS

The *Havhingsten's* oars measure 4.5m (15ft) long and their blades are 15cm (6in) wide. Research has shown that this is the most effective blade width for rowing long distances at sea.



▲ OAR PORTS

The ports were large enough for the oar blades to pass through. When the ship was under sail and the oars were not in use, the ports were sealed with special locks.



▲ SEATING FOR OARSMEN

Although the benches were narrow, they allowed the oarsmen to shift position regularly on long journeys. Sometimes the rowers simply sat on sea chests instead of benches.



▲ ROWING FOR THE SHORE

As they neared a hostile shore, Viking warriors fixed their shields to the gunwale to create an impressive display and to deflect spears and arrows.

◀ SHIP UNDER SAIL

On the open ocean, Viking sailors relied on a large, rectangular sail. When manoeuvring in coastal waters and up rivers, they dropped sail and rowed.